Visualization of β-reduction Graphs in the Lambda Calculus

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Introduktion

- Der findes visulaiseringer af lambda kalkyler
- Der findes ingen visualiseringer af reduktionsgrafer
- Forskere og undervisere

Design

- Todelt design
- Let at udvide
- Multiplatform (Python og GTK)

Grafgenerator

- Parser $\rightarrow$ Finde redexer $\rightarrow$ Reducer
- Generering af tilfldige grafer

Graftegner

- GraphViz
- Majoriazation
- PyGTK
- Python Widgets

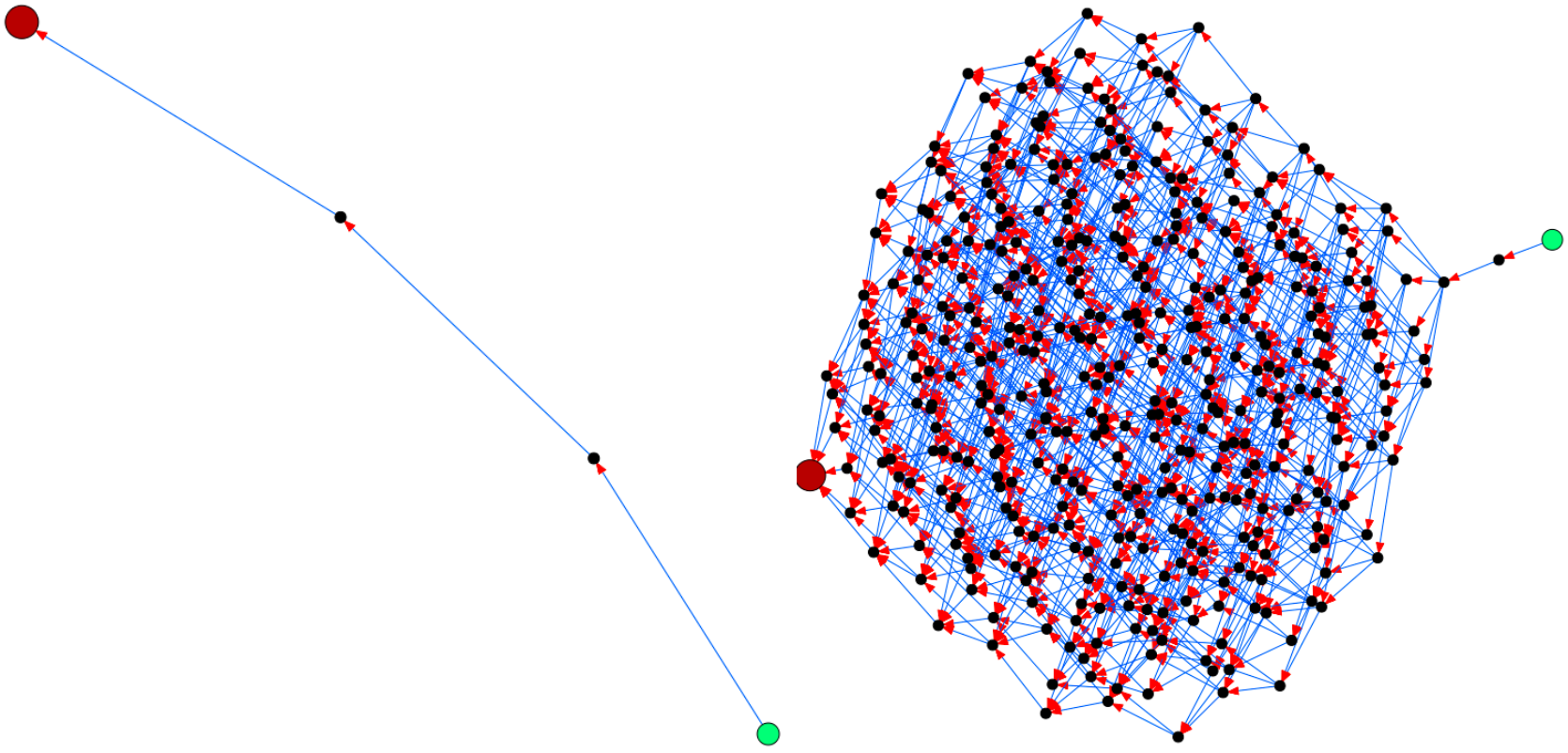
GUI

- Det er ikke en GUI opgave
- Enkel GUI

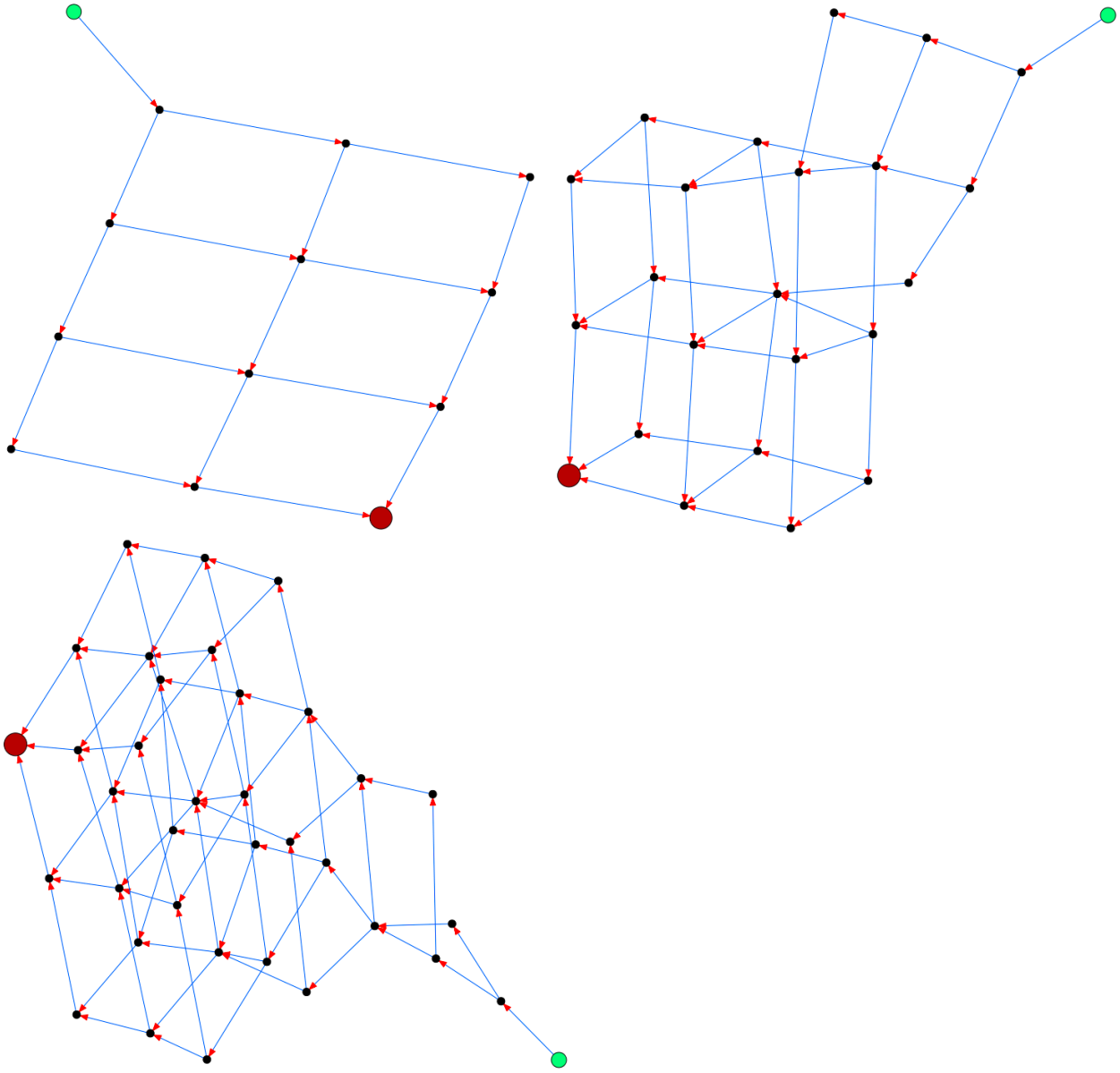
Eksempler

- Predecessor og Successor
- Addition, multiplikation og potens
- Hypercube
- Assosiation
- Cons

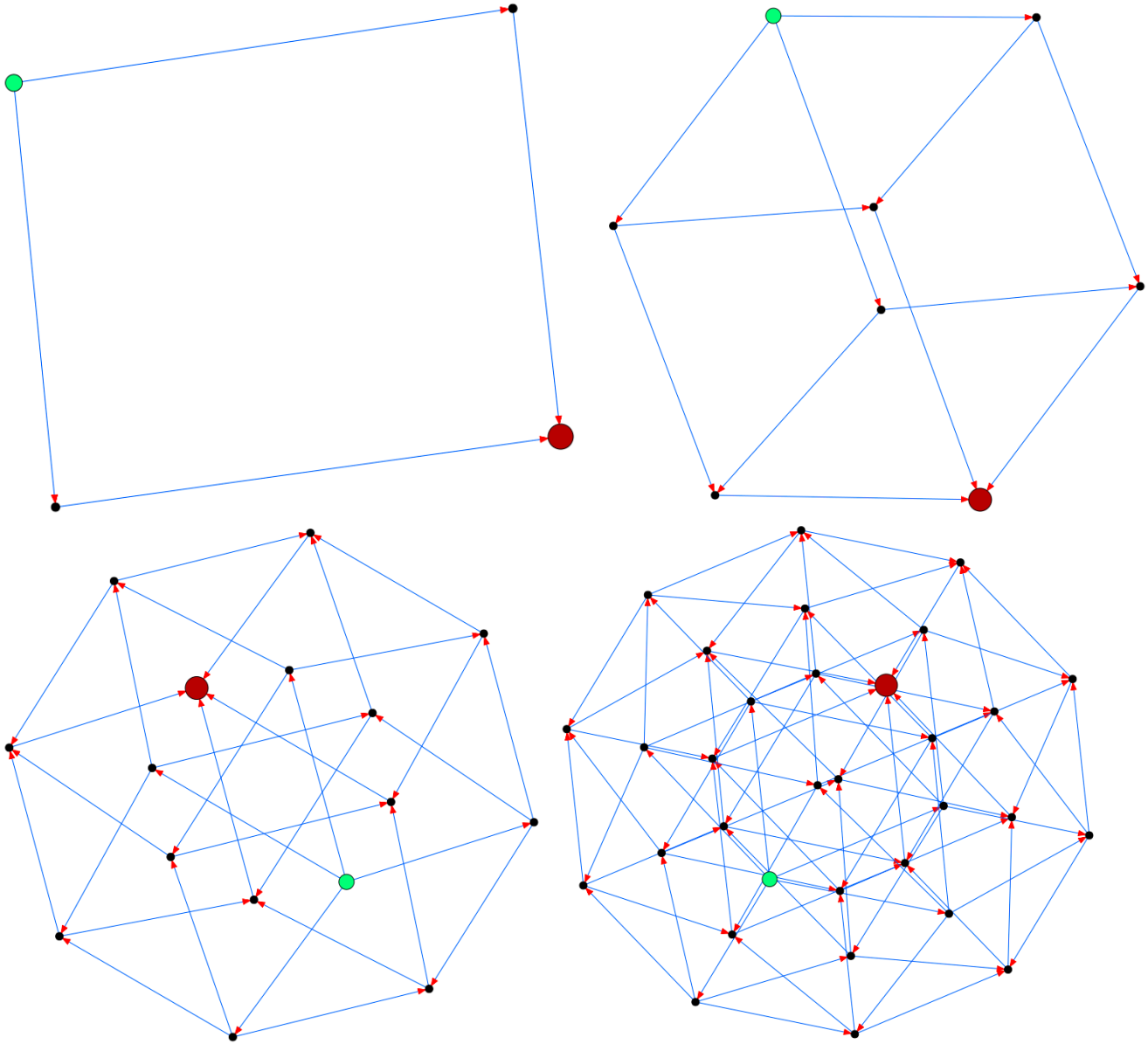
Predecessor og Successor



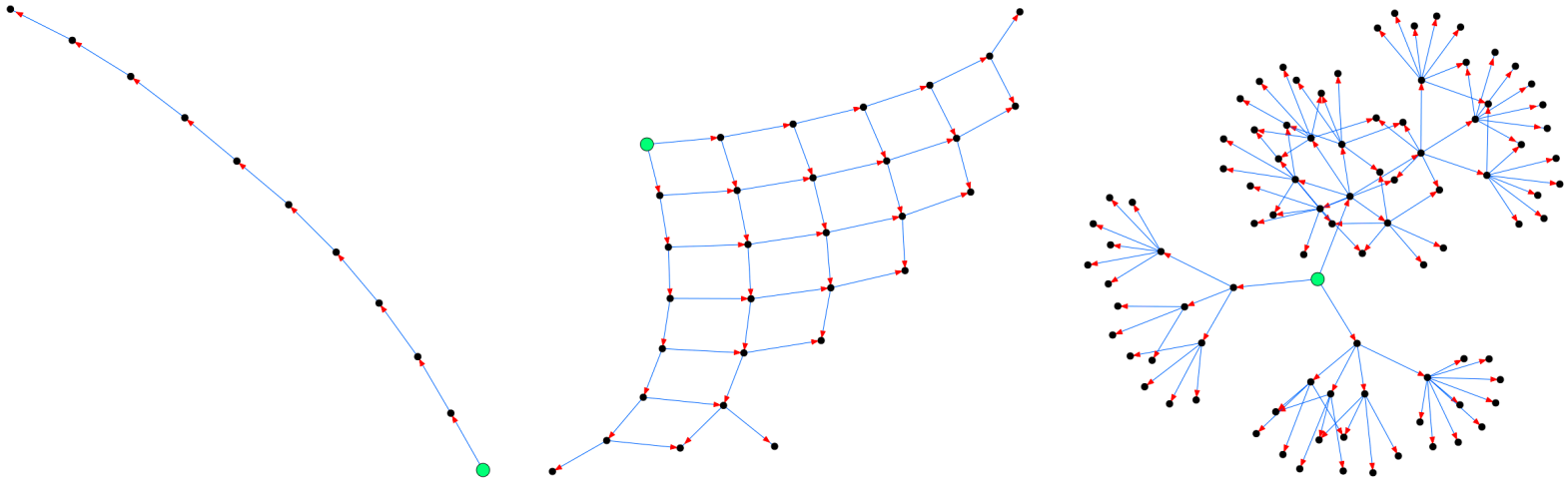
Addition, multiplikation og potens



Hypercube $(((\lambda x_1.x_1)f)\ldots((\lambda x_n.x_n)f))$

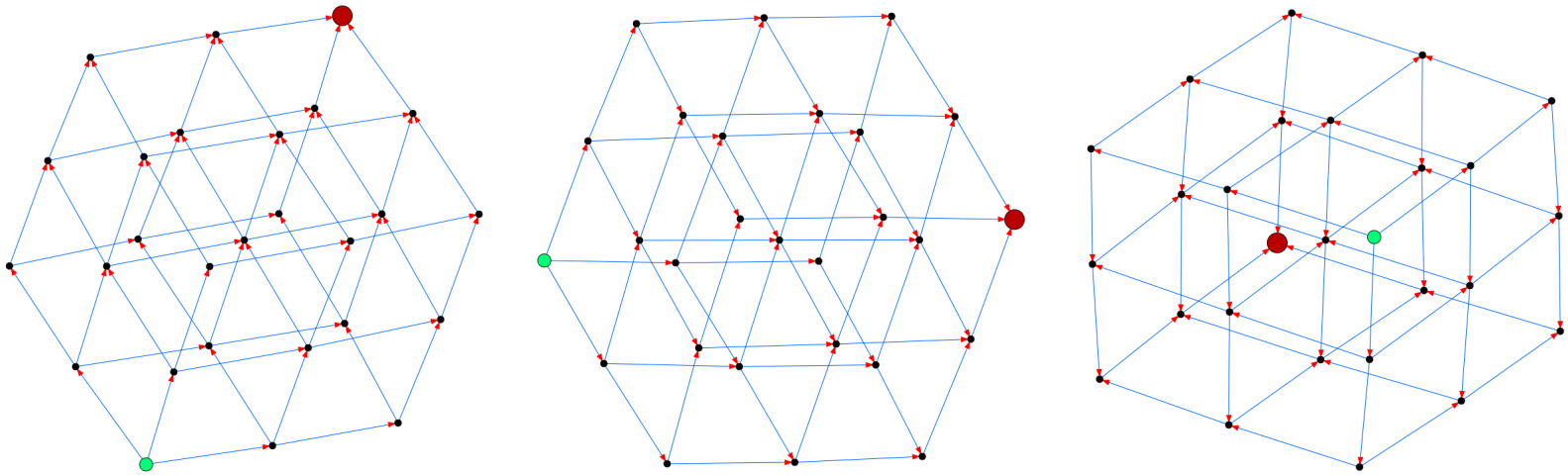


Assosiation



(a) $\omega_3\omega_3\omega_3\omega_3$ (b) $\omega_3\omega_3(\omega_3\omega_3)$ (c) $\omega_3(\omega_3(\omega_3\omega_3))$

Cons



(d) (2,3) (e) (4,4) (f) (5,0)

Figure: Cons funktionen

Performance

- Neato giver bedste all-round resultater
- Benchmarks

Fremtidigt arbejde

- Udbygge med flere graftegnealgoritmer
- OpenGL omskrivning (3D, zooming, etc.)
- Reduktionsstrategier
- Multitrdet
- Eksportering / Importering
- Reel brugertest

Konklusion

- En velfungerende 1. version
- Gode udbyggelsesmuligheder